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UX Research

Smart LMS with AI Proctoring

Survey Responses: 227 Participants

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1. Introduction

This report presents the findings of a user research survey conducted as part of the Empathize phase of our design thinking project. The survey targeted students who regularly use online learning and examination platforms, aiming to understand their pain points, frustrations, and desired features regarding online exams and academic platform management.

A total of 227 responses were collected, providing a robust and representative sample of the target user base. The insights gathered will inform the design of a smarter, fairer, and more integrated online proctoring and educational platform.

2. Research Objectives

The primary objectives of this research were to:

- Understand students' prior experience with AI-powered online proctoring systems.
- Identify the most significant pain points during traditional online exams (human-monitored via video call).
- Assess the impact of internet connectivity issues during online exams.
- Evaluate students' perception of fairness in current online examination systems.
- Discover the most common cheating methods that existing proctoring systems fail to detect.
- Determine desired features for an ideal educational platform.
- Understand the problem of platform fragmentation in students' academic lives.

3. Survey Findings

3.1 Prior Experience with AI Proctoring Systems

The survey opened by asking whether respondents had previously taken an online exam using an AI-based proctoring system.

Response	Percentage
Yes — I have used AI proctoring	50.7%
No — I have not used AI proctoring	49.3%

The near-even split (50.7% Yes vs. 49.3% No) indicates that AI proctoring is still in its adoption phase. Among those who had used such a system (120 respondents), 69.2% rated the experience as acceptable, 25% as excellent, and approximately 5.8% as poor. This suggests that while AI proctoring is functional, there is significant room for improvement to elevate user satisfaction.

3.2 Main Pain Points During Traditional Online Exams

Respondents were asked what bothers them most about traditional online exams conducted via live video platforms (e.g., Zoom) with a human proctor.

Pain Point	Percentage
Internet connectivity problems	54.6%
Being monitored by camera	24.2%
Psychological pressure / anxiety	21.1%

Internet connectivity is the dominant pain point by a wide margin (54.6%), underscoring the critical need for offline-tolerant or connection-resilient exam solutions. Camera monitoring (24.2%) and psychological pressure (21.1%) highlight the need for less invasive proctoring methods that reduce student stress.

3.3 Internet Disconnection During Exams

Students were asked how often they experienced internet disconnection during an online exam, causing anxiety about losing their answers.

Frequency	Percentage
Often	54.6%
Rarely	35.7%
Always / Never	~9.7% combined

A staggering 54.6% of students reported that internet disconnection during exams happens often. This is a critical finding, confirming that connectivity resilience is not a nice-to-have feature — it is an essential requirement.

When asked what a fair procedure would be in the event of disconnection, 68.7% of students preferred that the exam be paused temporarily until the connection is restored, while 25.6% preferred being allowed to finish and upload answers afterward. Only a small minority accepted immediate termination of the exam.

3.4 Perceived Fairness of Current Online Exams

Students were asked whether they feel that current online exams are fair and ensure equal opportunities for all students.

Perception	Percentage
No — cheating is easy and widespread	72.7%
To some extent	23.3%
Yes — exams are fair	~4%

An overwhelming 72.7% of students believe that current online exams are unfair because cheating is easy and widespread. This finding validates the core problem our platform aims to solve and highlights the urgent need for more effective and intelligent proctoring solutions.

3.5 Undetected Cheating Methods

Respondents identified cheating methods that they believe current proctoring systems cannot easily detect (multiple answers allowed).

Cheating Method	Responses (%)
Running AI tools in the background (e.g., ChatGPT)	174 (76.7%)
Having another person next to the screen	111 (48.9%)
Using an external monitor or projector	99 (43.6%)
Alt+Tab to open another browser tab	95 (41.9%)
Using screen-sharing/remote control (TeamViewer/Discord)	80 (35.2%)

The use of AI tools in the background is seen as the most difficult cheating method to detect (76.7%), followed by having someone present near the screen (48.9%). These results highlight the limitations of conventional proctoring and point to the need for behavioral analysis, AI-powered anomaly detection, and smarter environment monitoring.

Notably, 94.3% of students said they would very much appreciate being informed in real-time of any detected violation and the reason for it — indicating a strong preference for transparent, explainable proctoring systems.

3.6 Desired Platform Features

Students were asked which features they care most about having in an educational platform.

Desired Feature	Responses (%)
Progress tracking and performance monitoring	169 (74.4%)
Ease of use	161 (70.9%)
Clear and detailed exam performance reports	160 (70.5%)
Pre-exam reminders and notifications	120 (52.9%)

All four proposed features received strong support. Progress tracking (74.4%) and ease of use (70.9%) are top priorities, reflecting students' desire for a unified, intuitive platform that gives them visibility into their academic journey.

3.7 Platform Fragmentation Problem

Students were asked how many different platforms they are currently forced to use to follow a single academic subject.

Number of Platforms	Percentage
3 to 4 platforms	56.4%
More than 4 platforms	22%
2 platforms	17.2%
1 platform only	~4.4%

A striking 56.4% of students use 3 to 4 platforms per subject, while 22% use more than 4. This extreme fragmentation — juggling Google Classroom, Zoom, WhatsApp groups, Drive, university websites, and others — leads directly to missed deadlines and notifications. Indeed, 100% of respondents reported missing a deadline or lecture at least once because a link or notification was sent on a platform they weren't monitoring.

4. Empathize Phase

The Empathize phase is the first stage of the Design Thinking process. Its goal is to deeply understand the people we are designing for — their experiences, emotions, frustrations, and motivations. We used three core empathy tools: user interviews (via Google Form survey), an Empathy Map, and User Personas.

4.1 User Interviews (Google Form)

We conducted our user research using a Google Form survey distributed to university students. The form was designed to explore their direct experiences with online exams and educational platforms. It reached 227 respondents, providing a strong and statistically meaningful dataset.

Key themes explored in the interviews:

- Experience with AI-powered proctoring systems and satisfaction level.
- Biggest frustrations during live-proctored online exams (Zoom-style).
- Frequency and impact of internet disconnection during exams.
- Perceived fairness and integrity of current online exams.
- Awareness of cheating methods current systems fail to detect.
- Features students want most in an academic platform.
- The number of disconnected platforms students must juggle per subject.

4.2 Empathy Map

Based on the survey responses, we synthesized the following Empathy Map representing the collective voice of our target users — university students taking online exams.

EMPATHY MAP — Online Exam & Platform User	
💬 **SAYS** *"My internet dropped during the exam and I panicked about losing my answers."* *"Everyone cheats. I study hard but others use ChatGPT and nothing happens."* *"I missed the deadline because the link was on a platform I forgot to check."* *"The camera watching me the whole time makes me nervous even if I am not cheating."*	💭 **THINKS** The system punishes honest students while cheaters go undetected. A disconnection should never mean losing my work — that is not fair. I should not need 5 different apps to follow one subject. If I am flagged by the system, I deserve to know exactly why.
🎯 **DOES** Switches between 3–4+ platforms daily to follow up on lectures, assignments, and exams. Sits anxiously in front of the camera during exams, worried about connection drops. Misses deadlines because notifications are scattered across multiple apps. Refreshes the page repeatedly when the internet slows down during an exam.	❤️ **FEELS** Anxious and stressed during camera-monitored exams, even when not cheating. Frustrated and demotivated by a system perceived as unfair. Overwhelmed by the number of platforms and notifications to track. Hopeful — believes a smarter, unified, transparent system could genuinely help.
⚠️ **PAINS** Internet disconnections causing exam anxiety and potential answer loss. Perceived unfairness due to rampant undetected cheating (72.7%). Platform fragmentation leading to missed deadlines. Invasive camera monitoring causing psychological stress (24.2%).	🌟 **GAINS** A single platform that consolidates all academic activities and notifications. Smart AI proctoring that detects modern cheating without invasive monitoring. Transparent, real-time violation explanations (94.3% want this). Progress dashboards, clear exam reports, and timely reminders.

4.3 User Personas

Based on survey data patterns, we identified two primary user personas that represent the most common student archetypes in our target audience.

PERSONA 1 — The Stressed Honest Student

Name: Nour Ahmed | **Age:** 20 | **Role:** Sophomore, Computer Science

Background: Nour is a diligent student who prepares seriously for every exam. She has a moderate internet connection at home and has experienced disconnections during exams multiple times. She uses 4+ platforms per subject and frequently misses notifications.

Goals:

- Pass exams based on her own hard work without fear of unfair comparison.
- Track her academic progress clearly in one place.
- Never miss a deadline again.

Frustrations:

- Internet cuts out during exams, causing extreme anxiety about losing answers.
- Feels the system is unfair because cheaters are not caught.
- Overwhelmed by switching between WhatsApp, Google Classroom, Zoom, and the university portal.

Quote: *"I studied for days, but one internet drop during the exam and everything is at risk. That is not a test of knowledge — it is a test of your WiFi."*

PERSONA 2 — The Academically Overwhelmed Multitasker

Name: Karim Hassan | **Age:** 22 | **Role:** Senior, Business Administration

Background: Karim juggles a heavy course load and part-time work. He relies on notifications to stay on top of deadlines but finds the fragmented platform landscape impossible to manage. He has previously experienced AI proctoring and rated it as acceptable.

Goals:

- Manage all academic tasks and deadlines from a single, reliable platform.
- Receive timely, centralized notifications for exams and submissions.
- Understand his exam performance with clear, actionable reports.

Frustrations:

- Uses 4+ platforms per subject — misses important updates regularly.
- No visibility into overall academic performance or trends.
- Existing proctoring systems feel intrusive but still fail to deter obvious cheating.

Quote: *"I missed a submission because one professor used Google Classroom, another used a WhatsApp group, and a third posted on the university site. One platform for everything — please."*

5. Key Insights & User Needs

Based on the survey findings, the following key user needs have been identified:

- Connectivity resilience: Students need exams that handle internet drops gracefully, pausing rather than terminating sessions.
- Intelligent & unobtrusive proctoring: Less reliance on invasive camera monitoring; smarter AI-based behavioral analysis that detects modern cheating methods like background AI tool usage.
- Transparency: Real-time, explainable violation alerts so students understand what was flagged and why.
- Unified platform: A single, consolidated hub replacing the fragmented ecosystem of apps and websites.
- Progress visibility: Dashboards that track performance over time with clear exam reports.
- Timely notifications: Reliable reminders for exams and deadlines sent through a single, trusted channel.

6. Conclusion & Rationale for the Project Idea

The survey data strongly validates the need for a smarter online exam proctoring and educational management platform. The status quo is characterized by unfair exams (72.7% of students say cheating is rampant), technological fragmentation (78.4% use 3+ platforms per subject), and widespread connectivity anxiety (54.6% frequently experience disconnections).

Our project idea addresses these pain points by designing an integrated platform that combines AI-powered, non-invasive proctoring with a unified academic management experience — including progress tracking, clear reporting, and intelligent notifications. The research confirms that students are not only aware of these problems but are actively seeking solutions that are fair, transparent, and easy to use.

This empathy phase research provides a solid, data-driven foundation for moving into the Define and Ideate phases of the design thinking process.